
A Verifiable Environment for Neural Architecture Design

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Abstract

1 Reinforcement learning and agent evaluation both depend on a *verifier*: a function
2 that scores a candidate cheaply and without a human. Code and mathematics have
3 compilers and unit tests; neural architecture design has had no public verifier. We
4 present an environment in which an agent designs a neural network by emitting
5 edits to a typed, structured graph, and a deterministic, sub-millisecond verifier
6 grades the result against structural, budget, and serving-physics constraints – no
7 human or LLM in the scoring loop. The environment ships a procedural task gen-
8 erator (ten families, satisfiability and non-vacuity proofs, red-teamed anti-gaming
9 constraints), a difficulty-calibration harness, and a grounding study: across 264
10 architectures, verifier blockers predict PyTorch forward-pass failure at 96/96, and
11 we report honestly that the 0–100 health score is a validity margin, not a quality
12 ranking. Three findings follow. *As a repair signal*, one round of verifier feedback
13 lifts `claude-sonnet-4-6` from 79% to 100% and `deepseek-chat` from 59%
14 to 90%. *As a data foundry*, fine-tuning a 1.5B model on 3,000 minted verified
15 pairs lifts strictly-unseen pass@1 from 17% to 76%, and 100 GRPO steps on top
16 reach 80% ($n=192$); a contamination audit we ran on ourselves, disclose, and ship
17 shows the naive protocol would have reported 86%, quantifying the memorization
18 share. *As ground truth*, eleven LLM reward models score 0% false-positive on
19 blatantly broken designs yet approve 26–47% of near-miss designs one edit from
20 correct, failing exactly where reward hacking lives; the verifier is 0% on both
21 tiers. The same pure function is a leaderboard, an RL gym, and a verified-data
22 generator; we release all of it.

23 **Reproducibility note.** All results marked *measured* are reproduced by the commands in the repos-
24 itory; the calibration, amplification, and reward-model results use provider API keys, the rest need
25 neither keys nor GPUs. `VERIFICATION.md` in the repository maps each headline number to its
26 source artifact or reproduction command. Code and data: anonymized repository in the supplemen-
27 tary material; released under MIT upon publication.

28 1 Introduction

29 The scarce ingredient in modern agent training and evaluation is not the policy but the verifier. SWE-
30 Bench [1] grades a code patch by running the project’s own test suite; τ -bench [2] grades a tool-using
31 dialogue by diffing the resulting database state against a goal; in mathematics, programmatic answer
32 checking underpins both benchmark grading and verifier training [14]. None of these requires a
33 human judge or a second model, and this is precisely why they are trusted and why they can be used
34 as training signals.

Neural architecture design has lacked such a verifier, for a structural reason: whether a proposed network is even runnable (attention head counts that divide the embedding dimension, linear widths that match upstream shapes, a graph whose input reaches its output, a parameter count inside a budget) is a pure function only when architectures are represented as typed, structured graphs rather than free-form code. Coding agents emit code, not graphs, so their output carries no shape-level verifiability. We close this gap by building the environment around a typed graph and the deterministic checks that a production editor already runs on it. Existing structured views of architectures, such as Netron [13], are read-only viewers: they render a graph but do not edit it, propagate shapes, or verify, and so cannot serve as an action space or a reward.

Our contributions are: (1) a design-from-spec and edit-in-place environment with a sub-millisecond programmatic verifier; (2) a procedural task generator with satisfiability proofs, non-vacuity proofs, and anti-gaming constraints; (3) a calibration harness that measures difficulty against the labs’ usable band, with keyless self-tests that bracket the harness itself; (4) a grounding study connecting verdicts to real PyTorch behavior, reported with its negative results intact; and (5) a set of empirical results released with the environment: a verifier-in-the-loop lift ($79 \rightarrow 100$, one repair round, replicated on a second model), a strictly-held-out lift from 17% to 76% by SFT on environment-minted verified pairs and to 80% with GRPO on top ($n=192$) (with a disclosed contamination audit that quantifies the memorization share of the naive protocol, and the RL-from-raw null reported and explained), and an LLM-reward-model audit whose 0% false-positive rate on blatant defects collapses to 26–47% on near-misses – atop a leaderboard, an RL gym, and a verified-data foundry.

2 The environment

State. A typed graph of a neural network: components drawn from the production editor’s vocabulary of 182 layer types, each carrying parameters, connected by directed edges; the released grader computes serving physics for the structurally load-bearing subset (linear, convolutional, embedding, normalization, and the attention family). **Action.** A batch of structured edits from a fixed vocabulary: `add_component`, `add_connection`, `update_params`, `delete_component`, `replace_model`, and others, the same vocabulary a production editor executes. **Reward.** A grading function returning a 0–100 health score plus a list of hard blockers; Figure 1 shows the full loop.

Formalization. The environment is a deterministic single-agent MDP $(\mathcal{S}, \mathcal{A}, T, R)$. A state $s \in \mathcal{S}$ is a typed graph (V, E) : nodes V each carry a type and a parameter dictionary, edges E are directed. An action $a \in \mathcal{A}$ is a batch of typed edits from the fixed vocabulary; the transition $T(s, a)$ applies them, with no environment stochasticity. The reward is the grader’s health score $R(s') \in [0, 100]$, and a task’s success predicate is $R(s') \geq \tau \wedge \neg \text{blocked}(s') \wedge \phi(s')$, where ϕ collects the task’s machine-checkable constraints (required types, parameter/KV/latency budgets, action-economy limits). Since T and R are pure functions, an episode is fully reproducible from its seed, and the same R is at once an evaluation metric, an RL reward, and a data-generation filter.

Widths, parameters, and cost. Each node type defines an output width w_{out} and a parameter count from its typed parameters. For the load-bearing types,

$$\text{linear}(d_{\text{in}}, d_{\text{out}}): \quad w_{\text{out}} = d_{\text{out}}, \quad |\theta| = d_{\text{in}} d_{\text{out}} + d_{\text{out}}, \quad (1)$$

$$\text{attention}(d, H): \quad w_{\text{out}} = d, \quad |\theta| = 4d^2 + 4d, \quad (2)$$

$$\text{conv}(c_{\text{in}}, c_{\text{out}}, k): \quad w_{\text{out}} = c_{\text{out}}, \quad |\theta| = c_{\text{in}} c_{\text{out}} k^2 + c_{\text{out}}. \quad (3)$$

Total parameters are $\text{PARAMS}(G) = \sum_{v \in V} |\theta(v)|$, and the decode-time KV-cache cost per token is

$$\text{KV}(G) = \sum_{v \in \text{attn}(G)} 2 H_v^{\text{KV}} \frac{d_v}{H_v} b, \quad (4)$$

for b bytes per cached value (multi-head attention is the case $H_v^{\text{KV}} = H_v$; grouped-query attention shrinks H_v^{KV}). A task with constraints ϕ and threshold τ is solved iff

$$\text{PASS}(T, G) = [B(G) = \emptyset] \wedge [h(G) \geq \tau] \wedge \phi(G), \quad (5)$$

where $B(G)$ is the hard-blocker set of Algorithm 1, $h(G)$ its health score (whose soft-penalty weights are implementation detail: every result in this paper depends only on the success predicate, and the score is diagnostic), and $\phi(G)$ conjoins the machine-checkable constraints: required

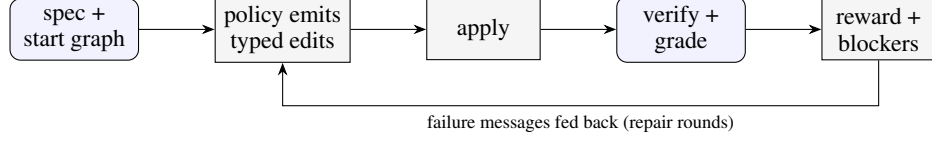


Figure 1: The environment loop. A policy edits a typed graph; a deterministic, sub-millisecond grader returns a reward and hard blockers. Feeding the blocker messages back for repair rounds is the amplification setting (Section 6); with no feedback it is single-shot.

Algorithm 1 $\text{GRADE}(T, G)$: the deterministic verifier

Require: task T with constraints ϕ and threshold τ ; typed graph $G = (V, E)$

Ensure: health score $h \in [0, 100]$; hard-blocker set B

```

1:  $B \leftarrow \emptyset$ 
2: if  $\neg \text{REACHES}(\text{in}(G), \text{out}(G))$  then  $B \leftarrow B \cup \{\text{DISCONNECTED}\}$ 
3: end if
4: for each attention node  $v \in V$  do
5:   if  $v.\text{embedDim} \bmod v.\text{numHeads} \neq 0$  then  $B \leftarrow B \cup \{\text{HEADDIV}(v)\}$ 
6:   end if
7:   if  $v.\text{numHeads} \bmod v.\text{numKVHeads} \neq 0$  then  $B \leftarrow B \cup \{\text{GQADIV}(v)\}$ 
8:   end if
9: end for
10: for each node  $v$  with typed input width  $w_{\text{in}}(v)$  and parent  $u$  do
11:   if  $w_{\text{in}}(v) \neq w_{\text{out}}(u)$  then  $B \leftarrow B \cup \{\text{SHAPEMISMATCH}(v)\}$ 
12:   end if
13: end for
14: compute  $\text{PARAMS}(G)$ ,  $\text{KV}(G)$ ,  $\text{LATENCY}(G)$ ; add a blocker for each budget in  $\phi$  violated, and
    for each required type absent
15:  $h \leftarrow 0$  if  $B \neq \emptyset$  else  $100 - \sum_{s \in \text{SOFT}(G)} \text{pen}(s)$ 
16: return  $(h, B)$ 
  
```

79 types, a two-sided parameter band $p_{\min} \leq \text{PARAMS}(G) \leq p_{\max}$, KV and latency budgets, and an
 80 action-economy limit $|a| \leq a_{\max}$. Every term is a pure function of G , so Eq. (5) is decidable in
 81 $O(|V| + |E|)$.

82 The grader composes three checks, each pure and sub-millisecond: a structural score with hard
 83 blockers (disconnected input/output, attention divisibility, parameter bloat), a guardrail pass over
 84 the proposed edits, and a shape propagator that catches shape bugs an edit would introduce. No
 85 LLM grades anything, so there is no persuasion surface and no stochasticity in the reward.

86 **The checks.** The grader runs three families of checks, in order. (i) *Structural blockers*: a dis-
 87 connected input or output; an attention layer whose `embedDim` is not divisible by `numHeads` (or
 88 `numHeads` not divisible by `numKVHeads` under grouped-query attention); a linear layer whose
 89 `inFeatures` do not match the upstream width; a parameter count far over budget. (ii) *Serving*
 90 *physics*: parameters, forward multiply-accumulates, and KV-cache bytes per token, computed with
 91 the canonical formula (grouped-query attention caches $2 \cdot \text{numKVHeads} \cdot d_{\text{head}} \cdot b$ bytes per token at
 92 b bytes per value; multi-head attention is the special case $\text{numKVHeads} = \text{numHeads}$), and checked
 93 against per-task parameter, KV, and decode-latency budgets. (iii) *Shape propagation*: a symbolic
 94 forward pass over the typed graph that catches interface mismatches an edit would introduce before
 95 any tensor is allocated. Each pass is $O(|V| + |E|)$ over the graph and completes in well under a
 96 millisecond, which is what makes the reward usable as an inner-loop RL signal rather than an offline
 97 evaluation.

98 A task pairs a natural-language specification with a starting graph and machine-checkable con-
 99 straints, e.g. “*This encoder fails validation: embedDim (192) is not divisible by numHeads (7).*
 100 *Repair the attention configuration in place with at most 2 actions. Do not rebuild the model from*
 101 *scratch.*” Figure 2 shows this state and the blocker the verifier returns.

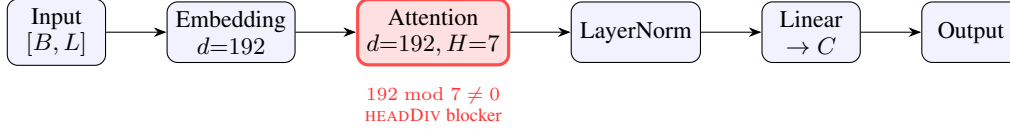


Figure 2: A task state as a typed graph, and a hard blocker the verifier returns in $O(1)$ at the attention node: embedDim=192 is not divisible by numHeads=7 (Algorithm 1, line 5). Because widths and divisibility are typed-node properties, this check is a pure function; on free-form code it would require executing the model. The input carries shape $[B, L]$ (B batch, L sequence length), widening to $[B, L, 192]$ after the embedding. The repair task asks the policy to fix exactly this in ≤ 2 edits without rebuilding.

Failure category (curated split, <code>claude-sonnet-4-6</code>)	count
missing required layer type (e.g. attention)	3
disconnected graph (input does not reach output)	2
health score below threshold	1
over parameter budget	1
passed	7 / 12

Table 1: First-try failure reasons of a frontier model on the curated production-architecture split: 5 of 12 tasks fail, and a failed task can trigger more than one detected reason (7 reasons across the 5 failures). Every reason is detected deterministically by the verifier.

3 Task generation

Curated tasks are a seed. A deterministic generator mints splits of any size from a seed across ten families: six design-from-spec (dense classifier, autoencoder, convolutional classifier, transformer encoder, grouped-query attention encoder, two-tower retrieval) and four edit-in-place (repair a broken attention configuration, trim an oversized model under a parameter budget, grow an undersized model into a two-sided parameter band, insert normalization). Because tasks are synthesized rather than scraped, a held-out split need never appear on the public web; contamination resistance against *model pretraining* is structural (unverified against a web index), and train/eval separation *within* the environment is enforced by an explicit holdout exclusion after our audit (Section 7).

Satisfiability and non-vacuity (measured). Every generated task carries its own reference solution. A 500-case sweep asserts that each reference passes its own task (no unsatisfiable tasks), and every edit-in-place start graph is asserted to fail its own task untouched (no vacuous tasks). Both run in CI without keys.

Anti-gaming. Edit-in-place families forbid `replace_model` and `clear_canvas`, so an agent that cannot diagnose a defect cannot pass by regenerating the whole network; budgets are two-sided bands rather than ceilings, so “delete everything” fails a trim task; a KV-cache budget is always paired with a required-attention constraint, so amputating attention cannot zero the cache. Nine such strategies, their defenses, and the pinned tests that fail if any defense is weakened are enumerated in the repository’s red-team report (`ROBUSTNESS.md`). An LLM-driven task proposer accepts nothing without a satisfiability proof: the proposer’s own reference must pass the grader under always-enforced safety-net constraints, so LLM authorship, unsafe for human- or LLM-judged benchmarks, is safe here.

A benchmark that bites (measured). On twelve hand-authored curated tasks built from real production architectures (a CIFAR CNN, a DLRM click-through-rate ranker, a Behavior Sequence Transformer, a two-tower retrieval encoder, a semantic-search encoder, and repair/scaling tasks), `claude-sonnet-4-6` passes 7/12 first try (mean health score 76); Appendix A sketches four of the tasks as typed graphs. The failures are machine-checkable and mundane (Table 1): it omits the required attention layer, leaves the graph disconnected, or blows the parameter budget by an order of magnitude. A benchmark a frontier model clears 12/12 measures nothing.

Family	single-shot	with verifier feedback
dense classifier	100%	100%
autoencoder	100%	100%
convolutional classifier	83%	100%
GQA encoder	100%	100%
repair attention	100%	100%
trim under budget	100%	100%
grow to band	100%	100%
two-tower retrieval	100%	100%
transformer encoder	8.3%	100%
insert normalization	0%	100%
overall	79%	100%

Table 2: Per-family pass rate for `claude-sonnet-4-6` on the public benchmark (generator v2; $n=12$ per family, all 120 tasks graded with zero exclusions). Seven families sit at 100% and the convolutional classifier at 83%; the two the model finds hard are exactly where verifier-in-the-loop feedback (second column; Section 6) pays off. Overall single-shot 79% (Wilson 95% CI [71, 85]) rises to 100% ([97, 100]); the v1-generator measurement (82% $\rightarrow$ 100%, $n=117$) replicates within three points.

131 4 Difficulty calibration

132 An ungameable environment is still useless at 0% pass (no learning signal) or near 100% (nothing to
133 learn). Practitioners cite a floor near 2–3% pass rate on the hard end (roughly one success per 33–50
134 rollouts) for a usable RL environment. The calibration harness measures per-family pass rates with
135 Wilson 95% intervals [15] and flags each family’s band. Two keyless self-test policies bracket the
136 harness itself: a reference policy that replays known-good solutions (measured: 100% pass) and
137 a noop policy that submits empty plans (measured: 0% pass). If either self-test deviates, the harness
138 rather than the model is broken, and CI fails.

139 **Measured** (`claude-sonnet-4-6`, **public benchmark**, $n = 12/\text{family}$). Single-shot pass rate
140 per family: most families saturate at or near 100% (a curriculum tier for a frontier model) and
141 two are hard: transformer encoder (8.3%) and insert-norm (0%) (Table 2). Deterministic grading,
142 reproducible from the seed.

143 5 Grounding study

144 Does the verifier’s verdict correspond to reality? We generated clean reference architectures and
145 systematically corrupted variants (broken attention divisibility, mismatched linear width, a severed
146 mid-graph edge), built every graph as a PyTorch module, and trained each briefly on a synthetic
147 fixed-teacher task.

148 **Result (measured, 264 graphs, two seeds).** The 264 graphs comprise three classes: 80 clean
149 references, 96 the verifier blocked, and 88 deliberately width-corrupted. Verifier blockers predicted
150 a forward-pass failure with perfect precision: 96 of 96 blocked graphs failed to run. The clean
151 graphs all constructed (80 of 80) and made training progress in 90%. The third class quantifies the
152 transparent rubric’s one systematic miss: it does not chase linear widths through the graph, so all 88
153 width-corrupted graphs pass the rubric yet fail the forward pass (88/88); the richer verifier in the
154 production system flags this class. We therefore claim the blocked side (a blocker proves runtime
155 failure) and report the blind spot as a measured limitation rather than hiding it (Figure 3).

156 **Negative result (measured).** Within clean graphs, the 0–100 score’s magnitude does *not* rank
157 training quality: Spearman correlation between the score and relative loss decrease was -0.58 ,
158 because deeper, more structured models make slower per-step progress on the short synthetic probe
159 than tiny ones. The score is therefore a validity margin, not a quality ranking. A learned quality
160 head, trained on (architecture, verdict, real training curve) triples that the environment can mint at

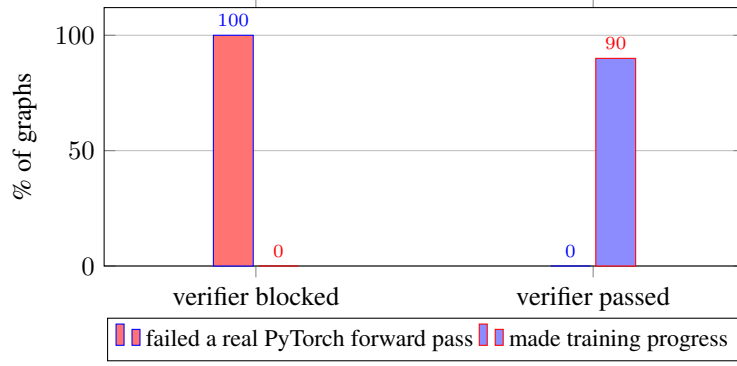


Figure 3: Grounding study (264 graphs, two seeds: 80 clean, 96 blocked, 88 width-corrupted). Every graph the verifier blocked (96/96) failed a real PyTorch forward pass; every clean graph passed and constructed (80/80), with 90% making training progress. The width-corrupted class (not shown) passes the transparent rubric yet fails 88/88 forward passes – the rubric’s blind spot, measured in the text.

Algorithm 2 ROLLOUT(T, π, k): verifier-in-the-loop repair

Require: task T ; policy π ; maximum repair rounds k

Ensure: pass $\in \{0, 1\}$; rounds used

```

1:  $G \leftarrow T.\text{startGraph}$ ;  $m \leftarrow \emptyset$  ▷  $m$ : verifier feedback
2: for  $t \leftarrow 1$  to  $k$  do
3:    $a \leftarrow \pi(T.\text{spec}, \text{SERIALIZE}(G), m)$  ▷ batch of typed edits
4:    $G \leftarrow \text{APPLY}(G, a)$ 
5:    $(h, B) \leftarrow \text{GRADE}(T, G)$ 
6:   if  $B = \emptyset \wedge h \geq \tau \wedge \phi(G)$  then return  $(1, t)$ 
7:   end if
8:    $m \leftarrow \text{EXPLAIN}(B)$  ▷ the verifier’s failure messages
9: end for
10: return  $(0, k)$ 

```

161 scale, is the natural next step and the point at which the verifier would become a calibrated predictor
 162 rather than a gate.

163 **6 Verifier-in-the-loop amplification**

164 Single-shot is the $k=1$ special case (no feedback consumed); the amplification setting sets $k=3$.
 165 Algorithm 2 states the loop; the only difference between the two arms is whether line 8’s feedback
 166 m re-enters line 3. Because grading is free, the verifier’s failure messages can be fed back to a policy
 167 for repair rounds. We measure, per provider, the pass rate of a single shot versus the pass rate when
 168 up to k repair rounds are allowed, holding tasks, model, and prompt fixed so that the only variable is
 169 the feedback loop. The framing is deliberately pro-model: the environment’s value is the lift it adds
 170 to any model.

Model	single-shot pass	with verifier feedback	lift
claude-sonnet-4-6	79%	100%	+21 pts
deepseek-chat (V3)	59%	90%	+31 pts

Table 3: Verifier-in-the-loop lift, per provider. Same tasks, model, and prompt; the only variable is the feedback loop.

171 The lift is concentrated on the families a frontier model finds hard (Figure 4): of the 25 tasks
 172 feedback rescues, the transformer encoder (8.3% $\rightarrow$ 100%) accounts for 11 and insert-norm
 173 (0% $\rightarrow$ 100%) for 12, with the convolutional classifier (83% $\rightarrow$ 100%) contributing the remain-

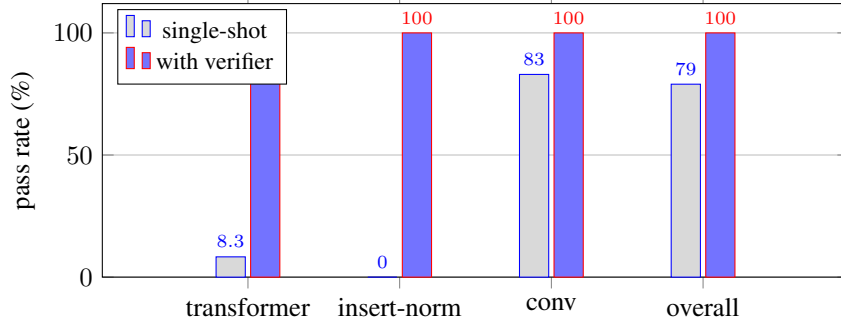


Figure 4: Verifier-in-the-loop lift for `claude-sonnet-4-6`. The overall gain (79 $\rightarrow$ 100; 25 tasks rescued) is concentrated on the two families the model finds hard (11 fixes on the transformer encoder, 12 on insert-norm), with the convolutional classifier contributing the remaining 2; the seven families already at 100% single-shot are omitted.

max repair rounds k	claude-sonnet-4-6	deepseek-chat
$k = 1$ (single-shot)	79% [71, 85]	59% [47, 71]
$k = 2$	100% [97, 100]	90% [80, 95]
$k = 3$	100% [97, 100]	90% [80, 95]

Table 4: Ablation on the number of verifier-feedback repair rounds, two models (generator v2; claude $n=120$, deepseek $n=59$; Wilson intervals). For *both*, a single repair round ($k=2$) provides the entire lift – every task the verifier can help fix is fixed after one round of its failure messages – and $k=3$ adds nothing. The pattern is not model-specific.

ing 2; the seven other families are already at 100% single-shot. A single repair round closes the whole gap (Table 4). Numbers are from generator v2 with all 120 tasks graded, and they *replicate* the v1 measurement (82% $\rightarrow$ 100%, $n=117$) within three points. The effect is not model-specific: on the same tasks, a second, weaker model (deepseek-chat) goes 59% $\rightarrow$ 90% (+31 points, $n=59$; v1: 60 $\rightarrow$ 88), with every fix again landing at $k=2$ and none at $k=3$. The weaker the model, the more the verifier is worth.

7 Training on the environment

The environment trains models through two channels, and we report both honestly.

Verified SFT: the data foundry works. Every generated task carries a reference solution that provably passes the grader, so the environment mints unlimited verified (spec $\rightarrow$ edits) supervised pairs, keylessly and for free. Fine-tuning Qwen2.5-1.5B-Instruct [17] on 3,010 such pairs (LoRA [16], one free T4, under ten minutes), with every training row verified to share no task with the evaluation split, lifts strictly-held-out pass@1 from 17.2% (33/192, Wilson [13, 23]) to 76.0% (146/192, [70, 82]), and 100 GRPO steps atop the SFT checkpoint reach **80.2%** (154/192, [74, 85]; Table 5, Figure 5). An independent rerun of the GRPO stage from the same SFT checkpoint reproduces the headline exactly (154/192), while the secondary metrics move within run-to-run noise (parse failures 10 vs. 14, mean reward 1.12 vs. 1.10), so we treat the pass@1 gain – not the parse-failure reduction – as the replicated finding. The RL increment is directionally consistent across four independent runs (+9.4, +3.1, +4.2, +4.2 points), and the SFT level replicates across retrains (76.0% and 77.1% at $n=192$; 68.8% at $n=64$). The surviving failures are substantive design errors the verifier pinpoints (disconnected graphs, too few components), not formatting.

An audit, disclosed. These are the *corrected* numbers, and the correction is part of the contribution. Our own contamination audit found that the original sampler drew parameters from small discrete lists, yielding only 127 distinct tasks – so despite disjoint seeds, every task in the original eval split had appeared in training: the naive protocol measured 85.9% ([75, 92]) where the honest same-split number is 68.8% (44/64, [57, 79]). We widened the sampler to broad ranges (17,462 dis-

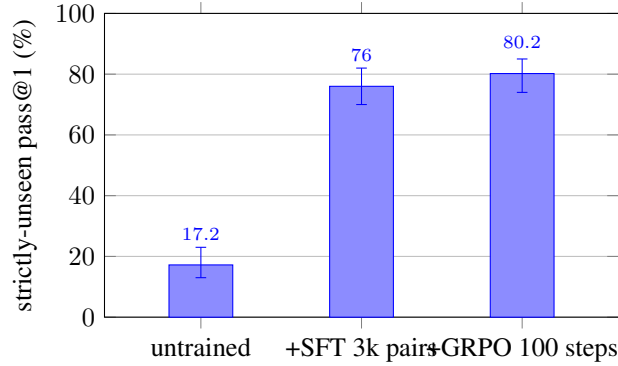


Figure 5: The corrected-protocol checkpoint chain (Qwen2.5-1.5B, seed-999 strictly-unseen split, $n=192$; error bars are Wilson 95% intervals). SFT on 3k environment-minted verified pairs lifts pass@1 from 17.2% to 76.0% with non-overlapping intervals; 100 GRPO steps on the merged SFT checkpoint reach 80.2%, a level an independent GRPO rerun reproduces exactly.

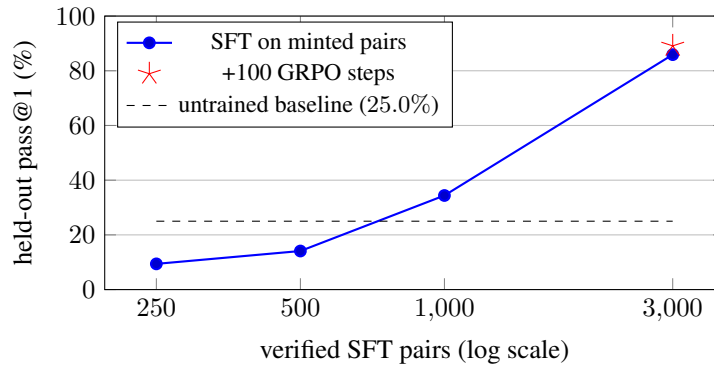


Figure 6: Held-out pass@1 vs. number of environment-minted verified SFT pairs (Qwen2.5-1.5B, LoRA, 2 epochs, seed-999 split, $n=64$). The curve is steep: below $\sim 1k$ pairs the half-learned format underperforms the raw instruct model; 3k pairs reach 85.9%. Pairs are free to mint, so the x-axis costs seconds of generation. The star marks 100 GRPO steps on the 3k checkpoint. Curve measured under the pre-audit v1 protocol (includes seen tasks); the corrected zero-overlap 3k numbers are 76.0% (SFT) and 80.2% (+GRPO) at $n=192$.

200 tinct tasks in a 20k draw at seed 999, reproducible keylessly; validated by the released satisfiability
 201 sweep), added an explicit holdout exclusion to the data minter (re-verified: 0 of the 192 evaluation
 202 tasks appear among the 3,010 training rows), and re-ran the pipeline. The 17-point gap between
 203 protocols is itself a measurement – the memorization share [18] of the naive number – and the audit
 204 script ships with the environment.

205 **Scaling, replication, and transfer.** Data scaling is steep and non-monotonic at the low end (Fig-
 206 ure 6, measured under the pre-audit protocol): 250 and 500 pairs *underperform* the raw instruct
 207 model (9.4% and 14.1% vs. 25.0%) – a half-learned edit format displaces the base model’s occa-
 208 sional valid outputs before replacing them – while 1,000 pairs recover to 34.4% and 3,000 reach
 209 that protocol’s ceiling. The lift replicates across independent sessions (85.9% and 79.7% under
 210 matching conditions). Transfer tracks data diversity, and the audit gave us a controlled comparison:
 211 models fine-tuned on the v1 data (127 distinct tasks) left the twelve hand-authored curated tasks
 212 unchanged (3/12, the untrained level), while the same recipe on the diverse v2 data doubles them
 213 to 6/12 ($n=12$, suggestive rather than conclusive; surviving failures are substantive, e.g. missing
 214 attention). Diversity is the operative lever for out-of-distribution transfer, and the generator makes it
 215 a parameter rather than a labeling effort.

	pass@1	parse failures	mean reward
<i>corrected protocol (v2, verified zero train/eval overlap, n=192)</i>			
Qwen2.5-1.5B (untrained)	17.2% [13, 23]	76/192	0.19
+ SFT on 3k clean pairs	76.0% [70, 82]	14/192	1.06
+ GRPO after SFT (100 steps)	80.2% [74, 85]	10/192	1.12
<i>pre-audit protocol (v1; eval tasks seen in training)</i>			
Qwen2.5-1.5B (untrained)	23.4% [15, 35]	19/64	0.35
+ SFT on 3k pairs	85.9% [75, 92]	2/64	1.19
+ GRPO after SFT	89.1% [79, 95]	0/64	1.26

Table 5: Training on environment-minted data (LoRA, one free T4; seed-999 splits, identical eval configuration; corrected block $n=192$, pre-audit block $n=64$). The corrected zero-overlap protocol is primary (one checkpoint chain, $n=192$; an $n=64$ run of the same protocol gives $14.1 \rightarrow 68.8 \rightarrow 71.9\%$, and an independent SFT retrain lands at 77.1%). The pre-audit block is retained because the gap between protocols measures the memorization share. Supervised fine-tuning on 3,000 verified pairs the generator mints for free lifts pass@1 by 59 points under the corrected protocol (62 pre-audit) with non-overlapping Wilson intervals and nearly eliminates parse failures. GRPO from the raw model is a null at this scale, but after SFT it works: the GRPO row trains from a fresh-session SFT replication (79.7%), within which RL adds +9.4 points and removes every remaining parse failure (see text). An independent rerun of the corrected-protocol GRPO stage reproduces its pass@1 exactly (154/192; parse 14, mean reward 1.10).

RL alone at small scale: an honest null, now explained. The released GRPO [6] loop runs end-to-end against the reward server on the same hardware, but RL *from the raw instruct model* does not move held-out metrics: two runs (learning rates $10^{-6} \times 150$ and $10^{-5} \times 250$ steps, every checkpoint evaluated and reported) sit at or below their own baselines. The SFT result explains the failure: the raw policy cannot emit parseable edits on a third of tasks, so the group-relative policy gradient starves. This is the classic SFT-then-RL split – supervised learning teaches the action format, RL then refines – and the environment supplies both stages: the verified SFT data and the deterministic reward. Completing the recipe confirms it: under the corrected protocol, 100 GRPO steps from the (merged) SFT checkpoint lift strictly-unseen pass@1 from 76.0% to **80.2%** ($n=192$), a gain an independent GRPO rerun from the same SFT checkpoint reproduces exactly (154/192 both times); mean reward rises $1.06 \rightarrow 1.12$ (1.10 on the rerun), while the parse-failure reduction ($14 \rightarrow 10$) does not replicate (14 on the rerun) and we do not claim it. The pre-audit chain showed the same pattern ($79.7 \rightarrow 89.1\%$). The same loop that was inert on the raw policy is productive once the policy can act, and every surviving failure is a substantive design error (disconnected two-tower graphs, component-count misses), not formatting.

8 Auditing LLM reward models with the verifier

Frontier labs increasingly use a strong model *as* a reward model to optimize objectives that are not directly verifiable (xAI reported this for Grok 4.1), a practice studied more broadly as LLM-as-a-judge [12], with process reward models [20] the analogous substitute where no programmatic check exists. The known failure mode is that an LLM reward model drifts: it approves answers that are actually broken. In a domain with a verifiable reward that drift is *measurable*. `reward_anchor.mjs` builds a verifier-labeled set (each design provably passes or fails: a reference design passes, its unsolved starting graph fails) and, given an LLM acting as a reward model, reports its agreement with the verifier and, crucially, its **false-positive rate**: how often it approves a design the verifier proves is broken. Across eleven reward models spanning closed-frontier (Claude, Grok) and open-weights (Qwen, Mistral, DeepSeek, Gemma, Llama), the false-positive rate is uniformly 0% (Table 7): not one approves a design the verifier proves broken. The failure mode that corrupts RLVR – rewarding a broken design – does not appear; the only errors are false *negatives* (1.7–13.3%, over-conservative), which waste valid designs but do not poison the reward. An LLM reward model here is therefore safe but lossy, and the verifier does the same job with 0% false negatives, deterministically and for free: a verdict costs microseconds of CPU, versus a multi-second, paid, stochastic API round-trip per judgment. This looks reassuring – but the negatives so far are blatant (an unsolved starting graph).

reward model	blatant FP	near-miss FP	near-miss FN
qwen-2.5-72b-instruct	0.0%	46.7% [35, 59]	1.7%
claude-sonnet-4-6	0.0%	33.3% [23, 46]	5.0%
grok-4	0.0%	25.5% [16, 39]	7.8%
deterministic verifier (ours)	0%	0%	0%

Table 6: The same audit with near-miss negatives on generator v2: passing designs one edit from correct ($n=60$ each; grok $n=51/37$ across tiers from provider 503s; corruption mix: dropped component, dropped connection, broken divisibility, each verified to fail). The v1 measurement (FP 22–48%) replicates. The 0% false-positive rate on blatant negatives collapses to 26–47%; the model that was perfect on blatant negatives (Qwen) is the worst here, approving whatever looks plausible. LLM judges fail precisely where reward hacking lives; the verifier does not.

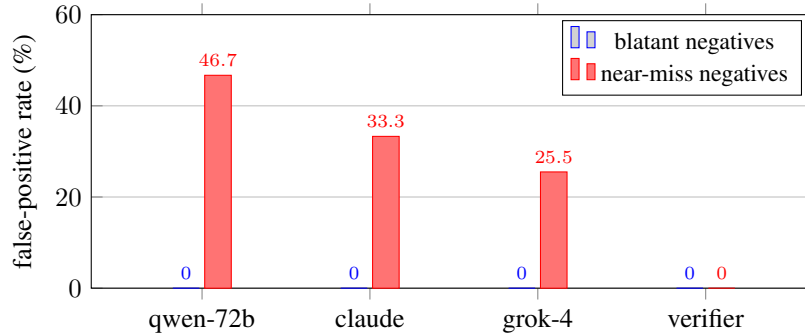


Figure 7: The collapse, visually: LLM reward models that never approve a blatantly broken design approve 26–47% of near-miss designs one edit from correct. The deterministic verifier is 0% on both tiers by construction.

The 0% does not survive near-misses. We re-run the audit with *near-miss* negatives: passing designs corrupted by a single edit (a dropped connection, an attention head count bumped off divisibility, a dropped component), each re-graded to confirm it fails. On these, the same judges collapse (Table 6, Figure 7): Claude approves 33% of provably-broken designs (Wilson [23, 46]), Grok 26% [16, 39], and Qwen-2.5-72B – among the strongest on blatant negatives (96.7% agreement) – approves 47% [35, 59] while rejecting almost nothing, i.e. it defaults to approval whenever a graph merely looks plausible. The inversion matters: LLM reward models are trustworthy exactly when errors are obvious, and fail exactly where reward hacking [19] lives – subtle defects one edit from correct. The deterministic verifier is 0% on both difficulty tiers by construction. This is a use of the environment beyond training or evaluation: a ground-truth anchor exposing when the LLM reward models labs deploy can and cannot be trusted.

9 Verified reasoning traces

The same verifier is a data foundry for the input reasoning-model post-training consumes: (spec $\rightarrow$ reasoning $\rightarrow$ design) triples where the design is re-graded and only passing traces are kept (rejection sampling). No trace enters the set unless its design provably solves the spec. A frontier model’s verified yield under this filter – the fraction of its reasoned designs that pass – was 72.5% (327 of 451 tasks, with 49 API failures excluded) on a 500-task run, producing 327 verified traces; the released tool records the yield per run and scales to larger sets. The 327-trace set is released publicly (dataset link withheld for double-blind review; included in the supplementary material). Because the split is seed-addressable, a private seed produces reasoning data whose designs never appeared on the public web.

model	tier	agreement	false-pos.	false-neg.
llama-3.3-70b-instruct	open	98.3%	0.0%	1.7%
qwen-2.5-72b-instruct	open	96.7%	0.0%	3.3%
claude-sonnet-4-6	frontier	95.0%	0.0%	5.0%
grok-4.5	frontier	95.0%	0.0%	5.0%
grok-4.20 (reasoning)	frontier	95.0%	0.0%	5.0%
grok-4-fast	frontier	95.0%	0.0%	5.0%
grok-4	frontier	94.6%	0.0%	5.4%
deepseek-r1 (reasoning)	open	93.3%	0.0%	6.7%
deepseek-chat (V3)	open	91.7%	0.0%	8.3%
mistral-large	open	88.3%	0.0%	11.7%
gemma-2-27b-it	open	86.7%	0.0%	13.3%
deterministic verifier (ours)	—	100%	0%	0%

Table 7: Eleven reward models vs. the verifier on generator v2 ($n=60$ each; grok-4 $n=37$ from provider errors). *Not one* has a false positive (0% across all); the only failure mode is over-conservatism (1.7–13.3% false negative). The three Grok variants agree with the verifier on the same 57/60 examples. The verifier matches the best model (0/0) at zero cost and deterministically. Section text and Table 6 show this 0% does not survive subtler, near-miss negatives.

	domain	verifiable reward	procedural, contam.-resist.	anti-gaming red-teamed	typed-graph action space
SWE-Bench [1]	code	✓	partial	—	—
τ -bench [2]	tool use	✓	—	—	—
SWE-Gym [3]	code	✓	partial	—	—
classical NAS [11]	architecture	proxy	—	—	—
ours	architecture	✓	✓	✓	✓

Table 8: Where this environment sits: architecture design as a verifiable-reward domain, with a contamination-resistant procedural generator, red-teamed anti-gaming defenses, and a typed-graph action space.

270 10 Related work

271 **Verifiable agent benchmarks.** SWE-Bench [1] and τ -bench [2] established the programmatic-
272 verifier pattern in code and tool use; SWE-Gym [3] showed that a few thousand verifiable tasks
273 suffice to train agents to a double-digit absolute improvement on SWE-Bench, and did so as free,
274 open research. We target the same pattern in a domain that lacked a public verifier (Table 8).

275 **RL environments and verifiable rewards.** Training on verifiable rewards (RLVR) is now central
276 to reasoning-model post-training [4, 5], typically with policy-gradient methods such as PPO [7] and
277 GRPO [6], preference optimization [9], or rejection sampling in the STaR [8] style. A growing
278 market supplies labs with RL environments; reward-hacking resistance is the quality bar most con-
279 sistently cited, and usable environments are expected to expose a hard band near a 2–3% pass floor.
280 We build directly against both criteria, and ship GRPO and STaR loops against the same grader.

281 **Neural architecture search.** Classical NAS [11], from RL-based search [21] to regularized evo-
282 lution [22], automates architecture design under a proxy objective, and NAS-Bench-101 [23] made
283 such search reproducible by exhaustively tabulating a fixed cell space; our environment differs in
284 providing a human-legible, per-edit, sub-millisecond verifier over a typed graph, and in serving as a
285 training and evaluation substrate rather than a single search procedure. The accompanying product
286 exposes a deterministic, constraint-satisfying design search (given a parameter budget, KV-cache
287 budget, and target GPU, it returns the Pareto front over capacity, parameters, and KV cache), a
288 NAS-lite special case built entirely from the verifier’s physics rather than a learned proxy objec-
289 tive. Differentiable and one-shot NAS [10] make gradient search tractable by relaxing a small,
290 hard-coded cell vocabulary and optimizing task loss; structural legality is assumed, not checked.
291 A per-edit legality verifier is orthogonal: it can veto the candidates of *any* search procedure – a
292 DARTS derivation, an evolutionary mutation, or an LLM agent’s edits – before GPU time is spent,
293 so it composes with these methods rather than competing with them.

11 Limitations

The transparent rubric released here does not verify linear-width consistency (Section 5); the health-score magnitude is not a quality ranking (Section 5); the reward path does not execute PyTorch per rollout, so shape-class failures are caught statistically rather than per-instance; and generated references lean on `replace_model` for design-from-spec families, biasing imitation style. Each is documented with its measurement or its planned mitigation in `ROBUSTNESS.md`. The training experiments use one small policy (Qwen2.5-1.5B, LoRA) with evaluation splits of $n=192$ (headline) and $n=64$ (replications); the SFT/RL gains are largely in-distribution (out-of-distribution transfer to the curated split is suggestive, $3/12 \rightarrow 6/12$ with diverse training data), and contamination resistance is argued structurally rather than verified against a web-scale index. The amplification study, the near-miss audit, and the eleven-model blatant-tier audit were all re-run on generator v2 (diverse instances) with results matching v1, resolving the earlier effective-sample-size concern for those measurements. Scaling the policy and diversifying the minted data further are the natural extensions.

12 Conclusion

Representing architectures as typed graphs turns “is this network sound, and what will it cost to serve” into a pure function, and this paper measures what that function buys. As a repair signal, one round of its feedback takes a frontier model from 79% to 100% and a weaker one from 59% to 90%. As a data foundry, the verified pairs it mints for free lift a small model from 17% to 80% on strictly-unseen tasks (SFT, then RL), under a train/eval-overlap audit we ran on ourselves, disclose, and ship. As ground truth, it exposes that LLM reward models with flawless records on blatant defects approve a quarter to a half of near-miss designs, failing precisely where reward hacking lives. We release the environment, the generator, the calibration and red-team harnesses, the training loops, and the public dataset, so that architecture design can join code and mathematics as a domain with a public verifier.

Broader impact

The environment lowers the cost of designing valid, servable architectures and of training and evaluating agents that do so, which should broaden access beyond teams with dedicated ML-research staff. Two risks warrant note. First, a verifier defines what “good” means; ours checks structural validity and serving cost, not fairness, safety, or data appropriateness, and must not be mistaken for a quality oracle – the negative result in Section 5 is deliberate. Second, verified-data generation can accelerate capability gains; we mitigate by releasing the methodology openly, so the same tools that build also audit (the reward-model analysis above). The benchmark contains no personal data and no scraped content: every task is synthetic or hand-authored.

Reproducibility

```
git clone <anonymized-repository>
cd arch-bench
npx vitest run                                # satisfiability + non-vacuity + anti-gaming
node calibrate.mjs --policy=reference          # harness self-test (must be 100%)
node calibrate.mjs --policy=noop              # harness self-test (must be 0%)
node training/build_sft_dataset.mjs \
  --count=3000 --seed=20260716                # mint verified pairs, holdout-excluded
python training/train_sft.py \
  --data arch-design-sft.chat.jsonl # 17.2% -> 76.0% strictly-unseen (T4)
python training/train_grpo.py --eval-only --seed 999 --count 192 \
  --model out/sft-arch/checkpoint-final
python training/train_grpo.py --steps 100 --lora --lr 1e-5 \
  --model out/sft-arch/checkpoint-final # SFT-then-RL -> 80.2%
node dump_grounding_set.mjs --count=40 --seed=123 --out=g.jsonl
python training/grounding_at_scale.py --set g.jsonl
```

python training/analyze_grounding.py

A Example curated tasks

The twelve curated tasks are hand-authored from real production architectures. Figure 8 sketches four, as the typed graphs the environment operates on; each is a distinct family (vision, ranking, sequence, retrieval) with its own required layer types and budgets.

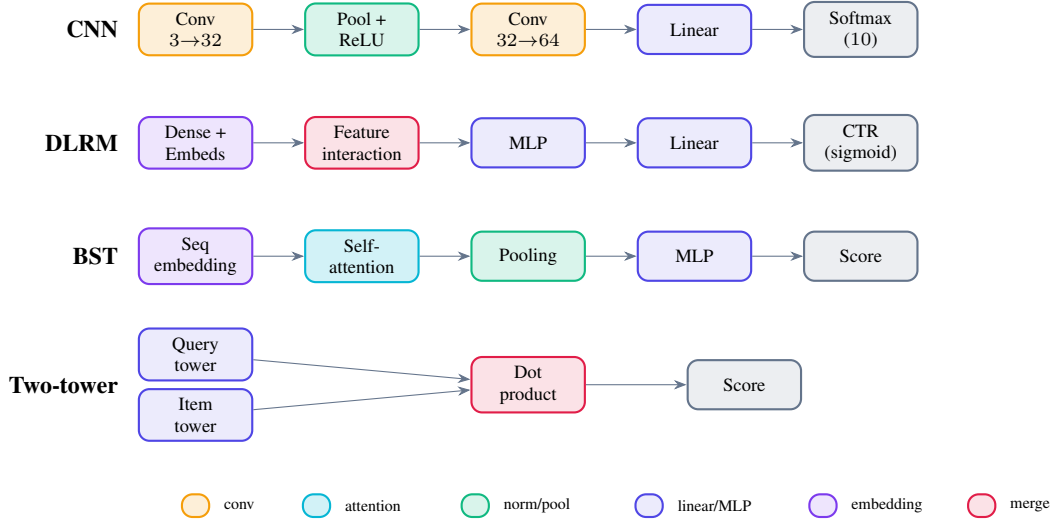


Figure 8: Four of the twelve curated tasks, as typed graphs, coloured by layer type (the same convention the accompanying production editor uses): a CIFAR-10 CNN, a DLRM click-through-rate ranker, a Behavior Sequence Transformer (which *requires* an attention layer), and a two-tower retrieval model (two encoders scored by a dot product). These are the states an agent edits and the verifier grades.

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